Class 10 Mathematics: Real Numbers Quick Revision & Formulas

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1. Fundamental Theorem of Arithmetic

Every composite number can be uniquely expressed (factorised) as a product of prime numbers, apart from the order in which the prime factors occur.
Formula: For any two positive integers a and b: **HCF(a, b) × LCM(a, b) = a × b**.

2. Proof of Irrationality

To prove that √2, √3, or √5 is irrational, we employ proof by contradiction. Assuming √2 = p/q in simplest coprime form leads to p and q both sharing a common factor 2, which contradicts the fundamental coprime assumption.

3. Decimal Expansions of Rational Numbers

Let x = p/q be a rational number such that prime factorisation of q is of the form $\mathbf{2^n \cdot 5^m}$, where n and m are non-negative integers. Then x has a decimal expansion which terminates. If q contains any other prime factor like 3 or 7, the decimal expansion is non-terminating repeating (recurring).

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